For the following problems, you need to explain your answer with a proper proof.

Problem 1. Let $H : a_1x_1 + \cdots + a_nx_n + a_{n+1} = 0$ be a hyperplane and let $\mathbf{v}(v_1,\ldots,v_n)$ be a vector which is not parallel to H. Denote by $\mathbf{a}$ the vector $(a_1,\ldots,a_n)$. Show that

$$\mathrm{Pr}_{H,\mathbf{v}}(P) = \left(I_n - \frac{\mathbf{v}\cdot\mathbf{a}^T}{\mathbf{v}^T\cdot\mathbf{a}}\right)\cdot P - \frac{a_{n+1}}{\mathbf{v}^T\cdot\mathbf{a}}\mathbf{v}.$$

Problem 2. Let π be a plane in $\mathbb{E}^3$ which passes through the origin and has normal vector $\mathbf{n}$. If $|\mathbf{n}| = 1$, show that

$$\mathrm{Pr}_{\pi}^{\perp}(P) = \left(I_3 - \mathbf{n}\otimes\mathbf{n}\right)\cdot P.$$

Problem 3. Let π be a plane in $\mathbb{E}^3$ which passes through the origin and has normal vector $\mathbf{n}$. If $|\mathbf{n}| = 1$, show that

$$\mathrm{Ref}_{\pi}^{\perp}(P) = \left(I_3 - 2\cdot\mathbf{n}\otimes\mathbf{n}\right)\cdot P.$$

Problem 4. Let $\phi(\mathbf{x}) = A\mathbf{x} + b$ be an affine transformation. Deduce the homogeneous matrix of the inverse transformation ϕ^{-1}.

Problem 5. Let π be a plane in $\mathbb{E}^3$ with normal vector $\mathbf{n}$ and equation $\pi : \langle\mathbf{n},\mathbf{x}\rangle = c$. Let $\mathbf{v}$ be a vector parallel to $\mathbf{n}$. Show that the composition of the orthogonal reflection in π followed by a translation by $\mathbf{v}$ is a reflection in the plane

$$\pi' : \langle\mathbf{n},\mathbf{x}\rangle = c + \frac{1}{2}\langle\mathbf{v},\mathbf{n}\rangle.$$

Problem 6. Show that an affine map $\phi(\mathbf{x}) = A\mathbf{x} + b$ is an isometry if and only if $A^{-1} = A^T$.

Problem 7. Show that a matrix A is in $\mathrm{SO}(2)$ if and only if

$$A = \begin{bmatrix}\cos(\theta) & -\sin(\theta)\\ \sin(\theta) & \cos(\theta)\end{bmatrix}$$

for some $\theta \in \mathbb{R}$.

Problem 8. Show that a direct isometry of $\mathbb{E}^2$ that fixes a point is either the identity or a rotation.

Problem 9. Let $\mathrm{Rot}_{\theta_1,C_1}$ and $\mathrm{Rot}_{\theta_2,C_2}$ be rotations with angles θ_1,θ_2 and centers C_1,C_2, respectively. Show that the composition of these two rotations is a (possibly trivial) translation if and only if $\theta_1 + \theta_2 = 2k\pi$ for some integer k.

Problem 10. Let $C(c_1,c_2)$ be a point. Deduce the homogeneous matrix of a rotation with angle θ and center C.

Problem 11. Show that a rotation with angle θ around the origin after a reflection in the x-axis is a reflection in the line $y = \tan(\theta/2)x$.

Problem 12. In dimension 2, discuss, with proof, the possible isometries obtained by composing a reflection in a line ℓ_1 with a reflection in a line ℓ_2 in terms of the relative positions of the two lines.

Problem 13. Show that a matrix $A \in \mathrm{SO}(3)$ admits 1 as an eigenvalue.

Problem 14. Show that a matrix $A \in \mathrm{O}(3)$ of determinant -1 admits -1 as an eigenvalue.

Problem 15. Let $\mathbf{v}$ be a unit vector and $\theta \in \mathbb{R}$. Show that the rotation with angle θ and axis $\mathbb{R}\mathbf{v}$ is given by

$$\mathrm{Rot}_{\mathbf{v},\theta}(\mathbf{x}) = \cos(\theta)\mathbf{x} + \sin(\theta)(\mathbf{v}\times\mathbf{x}) + (1-\cos(\theta))\langle\mathbf{v},\mathbf{x}\rangle\mathbf{v}.$$

Problem 16. Deduce the canonical equation of an ellipse.

Problem 17. For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, deduce equations for the tangent lines of a given slope.

Problem 18. For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, deduce an equation for the tangent line at the point (x_0,y_0) of the ellipse.

Problem 19. Show that the tangent lines to an ellipse at the endpoints of a diameter are parallel.

Problem 20. For the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, show that the tangent lines of a given slope k have equations

$$y = kx \pm \sqrt{a^2k^2 - b^2}.$$

Problem 21. For the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, deduce an equation for the tangent line at the point (x_0, y_0) of the hyperbola.

Problem 22. For the parabola $y^2 = 2px$ (where $p > 0$), show that the tangent line of a given slope k has the equation

$$y = kx + \frac{p}{2k}.$$

Problem 23. For the parabola $y^2 = 2px$ (where $p > 0$), deduce an equation for the tangent line at the point (x_0, y_0) of the parabola.

Problem 24. For the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, deduce an equation for the tangent plane at the point (x_0, y_0, z_0) of the ellipsoid.

Problem 25. For the cone $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$, deduce an equation for the tangent plane at the point (x_0, y_0, z_0) of the cone.

Problem 26. For the cone $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$, show that it is a ruled surface.

Problem 27. For the cone $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$, show that for any point P on the cone, the generator through that point is contained in the tangent plane at P to the cone.

Problem 28. For the hyperboloid $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$, deduce the equations of one of the families of generators.

Problem 29. For the hyperbolic paraboloid $\frac{x^2}{a} + \frac{y^2}{b} - 2z = 0$, deduce an equation for the tangent plane at the point (x_0, y_0, z_0) of the hyperbolic paraboloid.

Problem 30. For the hyperbolic paraboloid $\frac{x^2}{a} + \frac{y^2}{b} - 2z = 0$, deduce the equations of one of the families of generators.